

## Alps Alpine Resistive Position Sensor Board (Qwiic) - Application Notes

These are the application notes for the Alps Alpine Resistive Position Sensor Board (Qwiic) - Variants A & B.

### I2C Address Assignments (Variant A)

| A0 | A1 | I2C Address    |
|----|----|----------------|
| 0  | 0  | 0x5C (Default) |
| 0  | 1  | 0x5D           |
| 1  | 0  | 0x5E           |
| 1  | 1  | 0x5F           |

### I2C Address Assignments (Variant B)

| A0 | A1 | I2C Address    |
|----|----|----------------|
| 0  | 0  | 0x58 (Default) |
| 0  | 1  | 0x59           |
| 1  | 0  | 0x5A           |
| 1  | 1  | 0x5B           |

### Message Sequence & Format

All Alps Alpine QWIIC Sensor/HMI boards use a similar I<sup>2</sup>C communication sequence between the master and slave. In these exchanges, the Alps Alpine QWIIC Sensor/HMI boards always functions as the slave device.

The master is responsible for initiating both read and write operations. To change a setting or request data from a board, follow the sequence below:

**\*\* Placeholder Sequence. Add diagram later \*\***

- The Master must first send request message → The board will prepare the data or change setting as requested
- The Master must then send read message with size required for the response → The board will send back the requested data response

### Request Message Types

The following message types are used to indicate whether the board should prepare requested data for the user to read back or execute an operation. This message type ID is located at the beginning of the message payload sent from the I2C master device to the Alps Alpine Sensor/HMI board.

| Request type      | Message ID Value (uint8) | Description                                                                                                     |
|-------------------|--------------------------|-----------------------------------------------------------------------------------------------------------------|
| Request Data      | 0x01                     | Used to request the board to prepare data that is in the data request (proceeding value in the request payload) |
| Request Operation | 0x02                     | Used to request the board to execute a operation (proceeding operation in the request payload)                  |

### Request Data Types & Formats (0x01):

The following are different request data types that are supported by the Alps Alpine Resistive Position Sensor Board (Qwiic) and their expected formats.

#### “WHO AM I” Data Request Message (0x01):

The “WHO AM I” message can be used to both determine if a Alps Alpine Sensor/HMI board is present and can be used to determine what type of Sensor/HMI board it is.

Request I2C Payload:

| Byte Index | Value | Description           |
|------------|-------|-----------------------|
| 0          | 0x01  | Request Data type     |
| 1          | 0x01  | Who Am I Data Request |

Response I2C Payload:

During response request for “WHO AM I”, Master must perform a read request with **3 bytes**.

| Byte Index | Value                                                                                                                                 | Description                                                                                                             |
|------------|---------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| 0          | 0x01                                                                                                                                  | Message Header                                                                                                          |
| 1          | 0x01                                                                                                                                  | Message Header                                                                                                          |
| 2          | Resistive Position Sensor Type (uint8): <ul style="list-style-type: none"><li>• 0x06 = Variant A</li><li>• 0x0B = Variant B</li></ul> | Device type indicator.<br><br><b>Variant value of the requested Alps Alpine Resistive Position Sensor Board (Qwiic)</b> |

“Get Formatted Values” Data Request Message (0x02):

The “Get formatted values” message, can be used to request the current values from the Sensor board in a more friendly format for reading. Response data and its format may change based on current set modes. In the formatted request, the position value is presented the the user as a calculated angle value.

Request I2C Payload:

| Byte Index | Value | Description              |
|------------|-------|--------------------------|
| 0          | 0x01  | Request Data type        |
| 1          | 0x02  | Request Formatted Values |

Response I2C Payload:

In default operating mode, during the response request for “Get Formatted Values Data”, the Master must perform a read request with **2 bytes**.

| Byte Index | Content                                                             | Description                                                                  |
|------------|---------------------------------------------------------------------|------------------------------------------------------------------------------|
| 0 – 1      | <code>angleValue</code><br>( <code>uint16_t</code> , little-endian) | The angle position reading in degrees (Variant A 0–260) & (Variant B 0–320). |

|  |  |                                                                                                                                                                                                                          |
|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  | <p><b>** Note:</b> outside of supported angle range (Variant A between 261 - 360 degrees and Variant B between 320 - 360) odd value can be observed as this is out of the normal range of motion for this component.</p> |
|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### “Get Raw Values” Data Request Message (0x03):

The “Get Raw values” message, can be used to request the current raw data that the Sensor board reads from the Alps Alpine component. Response data and its format may change based on current set modes.

#### Request I2C Payload:

| Byte Index | Value | Description        |
|------------|-------|--------------------|
| 0          | 0x01  | Request Data type  |
| 1          | 0x03  | Request Raw Values |

#### Response I2C Payload:

In default operating mode, during the response request for “Get Raw Values Data”, the Master must perform a read request with **2 bytes**. The value presented represents the analog value read from the sensor component. The value thats provided in this request is the raw analog reading from the sensors output in.

| Byte Index | Content                                                   | Description                                                                                                                                                                                                                                                    |
|------------|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0 – 1      | <p>positionReading</p> <p>( uint16_t , little-endian)</p> | <p>The position reading (0–4095)</p> <p><b>** Note:</b> outside of supported angle range (Variant A between 261 - 360 degrees and Variant B between 320 - 360) odd values can be observed as this is out of the normal range of motion for this component.</p> |

#### **\*\* Calculation of Angle Value is different between Variant A & B:**

**Variant A:** `angleValue` (degrees) = ( `positionReading` / 4095) \* 260

**Variant B:** `angleValue` (degrees) = ( `positionReading` / 4095) \* 320

**“Get Is Data Ready” Data Request Message (0x04):**

The “Get Is Data Ready” message can be used to check whether the Sensor board is ready to provide component data readings when requested. It is good practice to perform this request after powering on the board and before requesting any data, continuing until the board indicates that it is ready. This message should also be used after any operation request to wait until the board is ready before sending additional requests.

**Request I2C Payload:**

| Byte Index | Value | Description                 |
|------------|-------|-----------------------------|
| 0          | 0x01  | Request Data type           |
| 1          | 0x04  | Request Is Data Ready Value |

**Response I2C Payload:**

In default operating mode, during the response request for “Get Is Data Ready”, the Master must perform a read request with 1 **byte**.

| Byte Index | Content                                              | Description                                                                                                       |
|------------|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| 0          | <code>isDataReady</code><br>( <code>uint8_t</code> ) | <ul style="list-style-type: none"><li>• 0x00 = Data <b>Not</b> Ready</li><li>• 0x01 = Data <b>Ready</b></li></ul> |

**“Get Firmware Version” Data Request Message (0x05):**

The “Get Firmware Version” message can be used to check the firmware version of the Sensor board.

**Request I2C Payload:**

| Byte Index | Value | Description              |
|------------|-------|--------------------------|
| 0          | 0x01  | Request Data type        |
| 1          | 0x05  | Request Firmware Version |

#### Response I2C Payload:

In default operating mode, during the response request for “Get Firmware Version”, the Master must perform a read request with **2 bytes**.

| Byte Index | Content                      | Description               |
|------------|------------------------------|---------------------------|
| 0          | MAJOR_VERSION<br>( uint8_t ) | The firmwares major value |
| 1          | MINOR_VERSION<br>( uint8_t ) | The firmwares minor value |

#### Request Operation Types & Formats (0x02):

The following are different request operation types that are supported by the Alps Alpine Resistive Position Sensor Board (Qwiic) and their expected formats. These requests do not have any responses in return.

##### “Enter DFU Mode” Operation Request Message (0xff):

The “Enter DFU Mode” message can be used to request the Sensor board to enter its DFU mode. This allows for a external I2C Master device to perform DFU over I2C interface.

#### Request I2C Payload:

| Byte Index | Value | Description            |
|------------|-------|------------------------|
| 0          | 0x02  | Request Operation type |
| 1          | 0xff  | Request Enter DFU Mode |